Trey Kirscht

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Summary

Electrical Engineering Co-op student specializing in embedded systems, hardware/firmware integration, mixed-signal prototyping, and bench validation. Experienced with low-level C and Python tooling, microcontroller firmware, PCB-based hardware, instrumentation, power/control systems, and test automation. Built and validated research and industrial prototypes spanning intraspinal stimulation hardware, cathodic protection rectifier control, waveform generation, and custom microcontroller programming tools.

Education

University of Alberta Expected 2026
Bachelor of Science in Electrical Engineering, Co-op *Edmonton, AB*

Technical Skills

Programming	C, C++, Python, MATLAB, VBA, JavaScript, ARM Assembly
Embedded	Microcontroller firmware, timers/interrupts, DMA, UART, SPI, I2C, ADC/DAC, hardware/software debugging
Hardware / PCB	Schematic capture, PCB layout, mixed-signal prototyping, power electronics fundamentals, analog measurement, soldering, board bring-up
Lab / Test	Oscilloscopes, multimeters, vector network analyzers, signal generators, logic analyzers, bench validation
Tools	Altium Designer, LTspice, PathWave ADS, Cadence Virtuoso, Ghidra, Git, Digilent WaveForms SDK

Engineering Experience

University of Alberta Research Lab 2026
Research Engineering Co-op / Hardware-Firmware Prototype Development *Edmonton, AB*

- Designed and validated a routed 48-channel intraspinal stimulation prototype using eight active stimulation lanes, analog current-source/sink output stages, crosspoint electrode routing, muxed measurement, and AD3 waveform control.
- Breadboarded and tested a one-channel mixed-signal output stage, validating biphasic current pulse behavior, command-capacitor timing, measurement-path behavior, and compliance limits under representative loads.
- Developed Python/WaveForms tooling for hardware-timed analog and digital waveform generation, pulse sequencing, measurement capture, calibration workflows, and test UI visualization.
- Designed the full prototype PCB, including power regulation, AD3 control interface, output stages, analog routing, measurement muxing, common-return switching, status indicators, and staged bring-up planning.

Seer Innovations Inc. Jan 2024 – Sep 2025
Engineering Support Specialist *Canada*

- Contributed firmware, testing, and integration work for a cathodic protection rectifier replacement project using controllable switch-mode power supplies, measurement, logging, remote monitoring, and operator tooling.
- Produced technical reports and engineering deliverables for inspection and corrosion-protection projects, including graphs, anomaly tables, data summaries, and supporting appendices.
- Built internal tools for survey data processing and CSV-to-KMZ generation to support field-use mapping deliverables.
- Maintained and improved a large VBA-based processing backend, supported workflow troubleshooting, and trained team members on tool usage.

Weir Concepts Jun 2022 – Sep 2023
Shop Technician *Edmonton, AB*

- Operated plasma cutting, CNC, and saw equipment to fabricate custom components from engineering drawings.
- Assembled and pressure-tested mechanical and hydraulic systems to API standards and supported shop material management.

Selected Projects

Cathodic Protection Rectifier Controller Unit 2025 – 2026
Capstone – Embedded / Firmware Lead

- Developed Particle Boron firmware for a retrofit rectifier controller with zero-cross synchronized SCR phase control, RMS/DC measurement, calibration, and closed-loop voltage control.

- Built cloud telemetry, diagnostics, waveform capture, history logging, and a Flask-based web UI for testing, validation, and control.
- Integrated hardware, firmware, and interface features for a working cathodic protection controller and supported debugging and verification in real rectifier hardware.

OpenAD3Flash 2026

Analog Discovery 3 Microcontroller Programming and Flash Tooling

- Built Python/WaveForms tools using a Digilent Analog Discovery 3 for AVR ISP programming, PIC16 high-voltage programming, EEPROM dumping, SPI flash dumping, and read-only glitch experiments.
- Implemented low-level bit-banged protocols, target-power control, signature reads, Intel HEX parsing, flash/verify workflows, and non-destructive read-only diagnostic tooling.
- Public repository: github.com/trey109231/OpenAD3Flash.

Embedded Waveform and Capture Tools 2024 – 2026

AVR / STM32 Firmware Projects

- Developed AVR C firmware for waveform generation with timer-driven output, external DAC control, LCD output, joystick/keypad input, and configurable waveform parameters.
- Built an STM32 timer/DMA waveform-generation and ADC-capture workflow with UART commands, circular DMA, binary capture export, and Python plotting.

Additional Technical Projects

Tiny AVR Mandelbrot Cluster 2026

Distributed Embedded Computing Experiment

- Built a five-node ATtiny84A Mandelbrot render cluster using a shared SPI-style bus, individual chip-select lines, fixed-point worker firmware, and Python host-side dynamic row scheduling.
- Public repository: github.com/trey109231/attiny84aMandelbrotCluster.

RF Amplifier Design and Test 2025

Course Project

- Designed a 2.3 GHz narrowband RF amplifier in PathWave ADS using Smith-chart matching, transmission-line modeling, S-parameter simulation, and PCB layout.
- Assembled and tested the fabricated PCB with a vector network analyzer and tuned the matching network to correct layout-dependent detuning.

Multilayer Device Fabrication and Characterization 2025

Course Project

- Fabricated and characterized a multilayer microdevice on silicon using PECVD silicon nitride deposition, aluminum sputtering, photolithography, wet etching, and electrical probing.